

Introduction

Why methane, why now. Methane traps $\sim 80\times$ more heat than CO_2 over 20 years and fades within a decade: fixing large leaks fast is one of the highest-leverage climate actions. U.S. EPA super-emitter notifications (implementation to Jan. 2027) identify any facility within **50 m** of an estimated leak, so 50 m is the scale at which localization becomes *actionable*.

The tools. A few inexpensive sensor masts watch the site; machine learning turns readings into a source estimate in milliseconds; split conformal prediction adds a guarantee: a region containing the true source 90% of the time.

The gap. (1) A coverage guarantee says how *often* the region is right, not how *small* it is, and no practical instrument measured how much smaller it could be. (2) Direct physics inversion needs the wind to be *known*; real sites only *measure* it.

Contributions.

C1 Physics input images that carry the wind sensor's error budget, attached to any network by a 0.2%-parameter head.

C2 An exact-posterior measuring stick: a calibrated deep localizer delivered 94–98 m where the data supported 6–7 m; the images close half the gap (46–48 m, three architectures).

C3 Under measured wind ($10^\circ/10\%$ error), where inversion returns the whole site, the method holds its guarantee at **70–73 m**.

Methods

Simulation with knowable truth.

Gaussian-plume transport (verified Briggs dispersion) on a 500 m site; readings $y = q g_c^*(w) + \epsilon$ are linear in the leak rate, so the *exact* probability map is computable under known wind: the smallest supported 90% region (6–7 m) audits every method in meters. Only the *measured* wind is recorded ($10^\circ/10\%$ per-step error); no model ever sees the truth.

Physics input images (closed form, milliseconds). With $a_c = g_c^\top y$ and $b_c = \|g_c\|^2$ computed under $K=8$ winds drawn from the sensor's error range: the *average evidence*, its *wind fragility*, the *sensitivity* $\log \bar{b}_c$, and the *fit quality*,

$$\bar{z}_c = \frac{1}{K} \sum_k \frac{a_c^{(k)}}{\sigma \sqrt{b_c^{(k)}}}, \quad v_c = \text{std}_k z_c^{(k)},$$

$$\bar{R}_c = \frac{1}{K} \sum_k (\rho - 2\hat{q}_c a_c^{(k)} + \hat{q}_c^2 b_c^{(k)}).$$

At $\epsilon=0$ these provably reduce to the classical matched filter ($R_c = \rho - \sigma^2 \max(z_c, 0)^2$): one recipe for exact and measured wind alike.

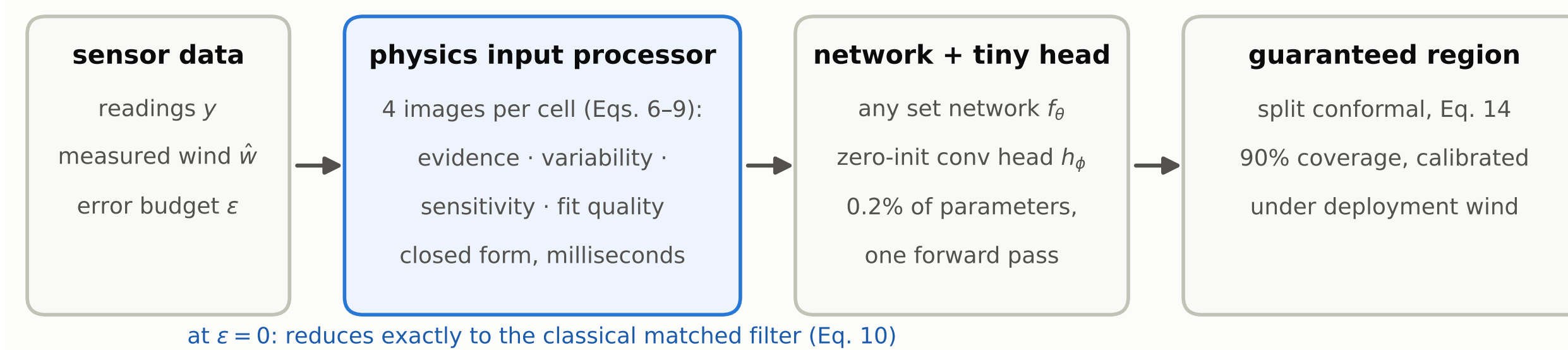
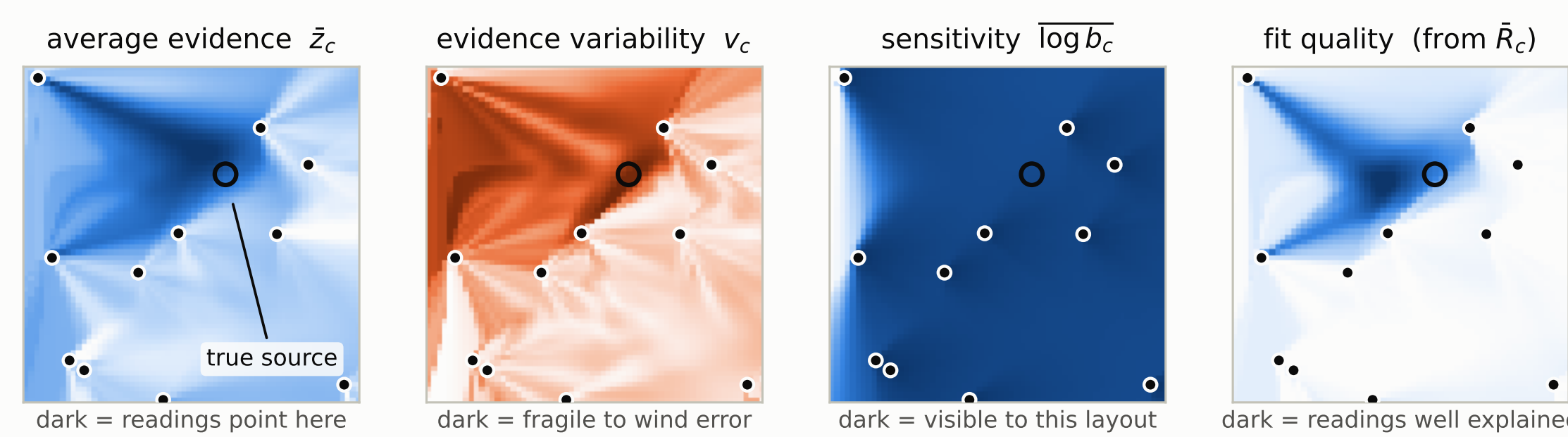
Learning and guarantee. Any set network f_θ plus a zero-initialized conv head h_ϕ reading the images, joined at the logits,

$$\ell(x) = f_\theta(x) + h_\phi(M(x)),$$

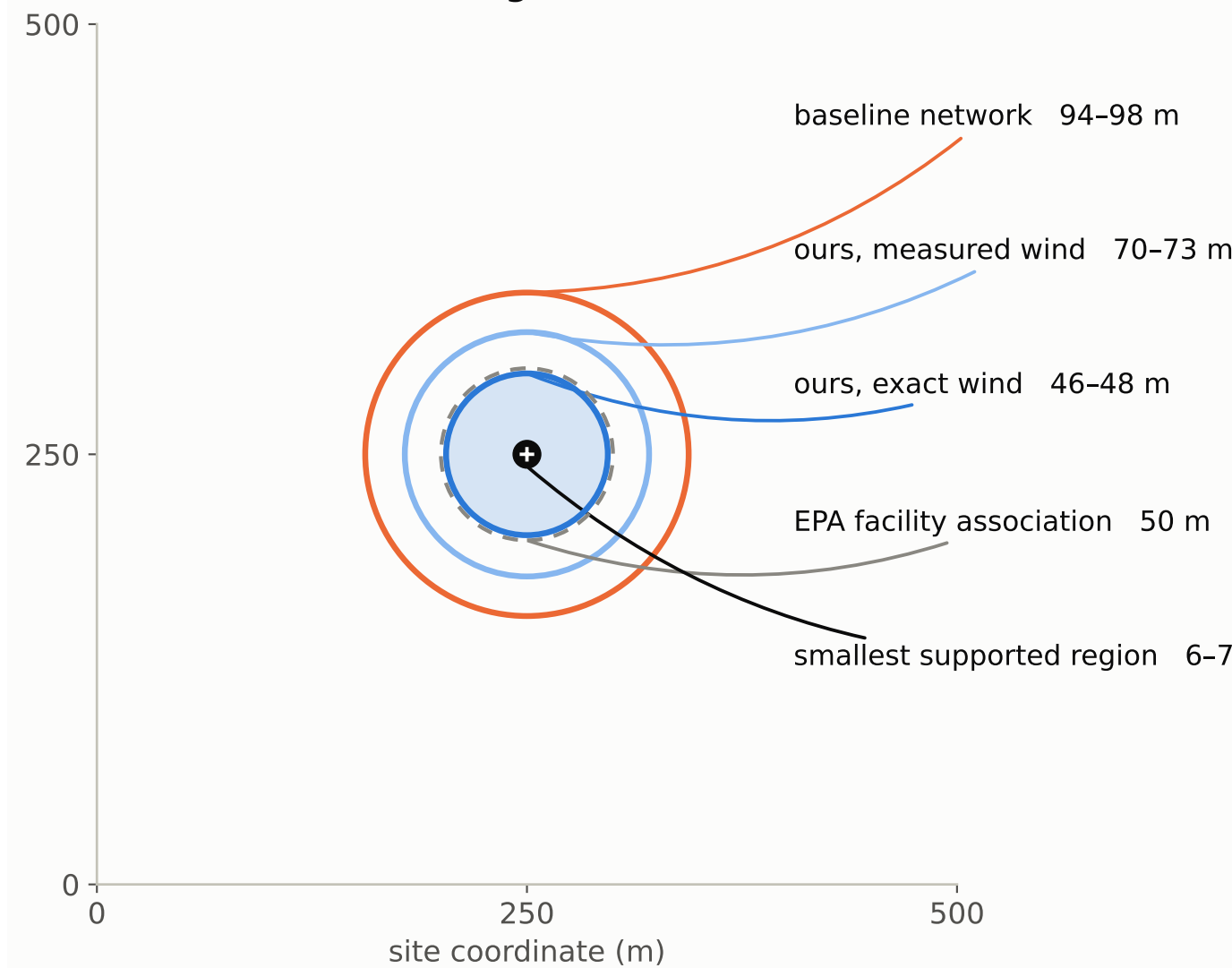
trained with ordinary labels; split conformal calibration under the deployment wind certifies $\Pr[c^* \in \mathcal{R}(x)] \geq 0.9$. Physics supplies exact evidence; learning supplies calibration and generalization.

Results

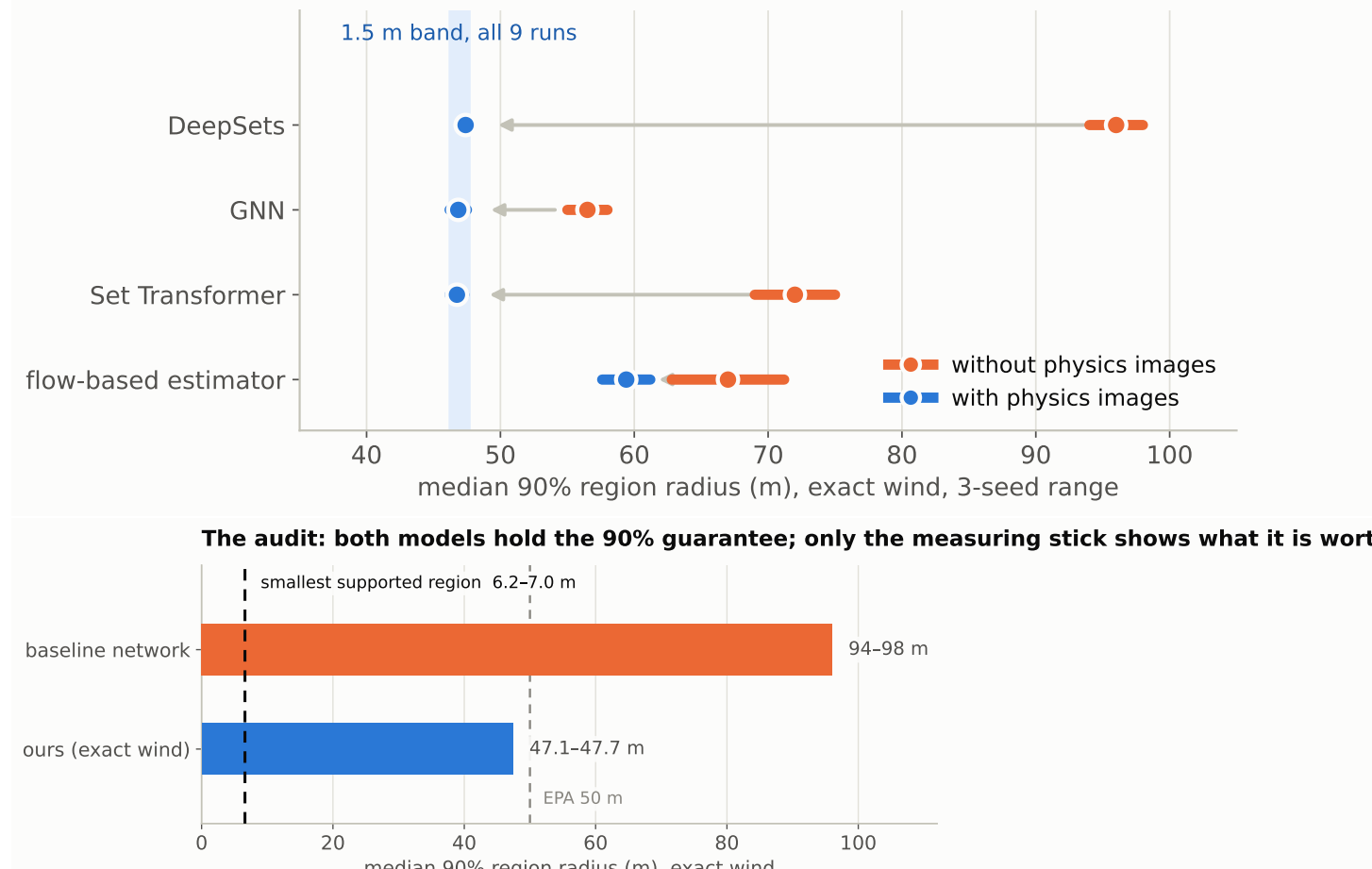
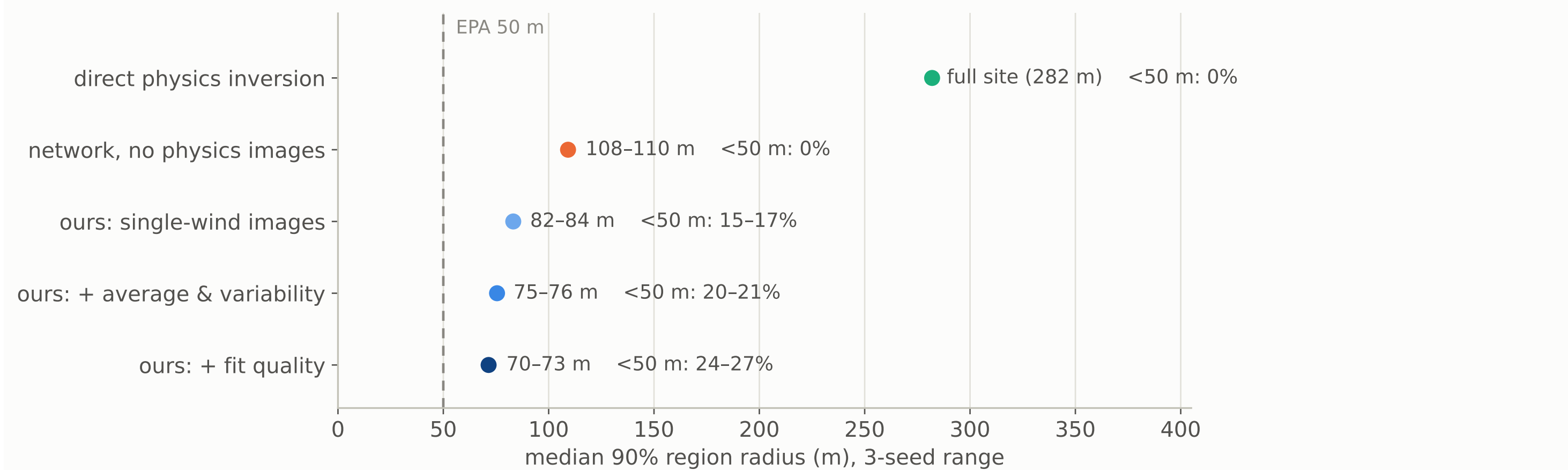
One budget-aware pipeline: physics supplies the evidence, learning supplies calibration and generalization

What the input processor computes for one scenario (closed form, milliseconds); $\circ$ marks the true source, dots the masts

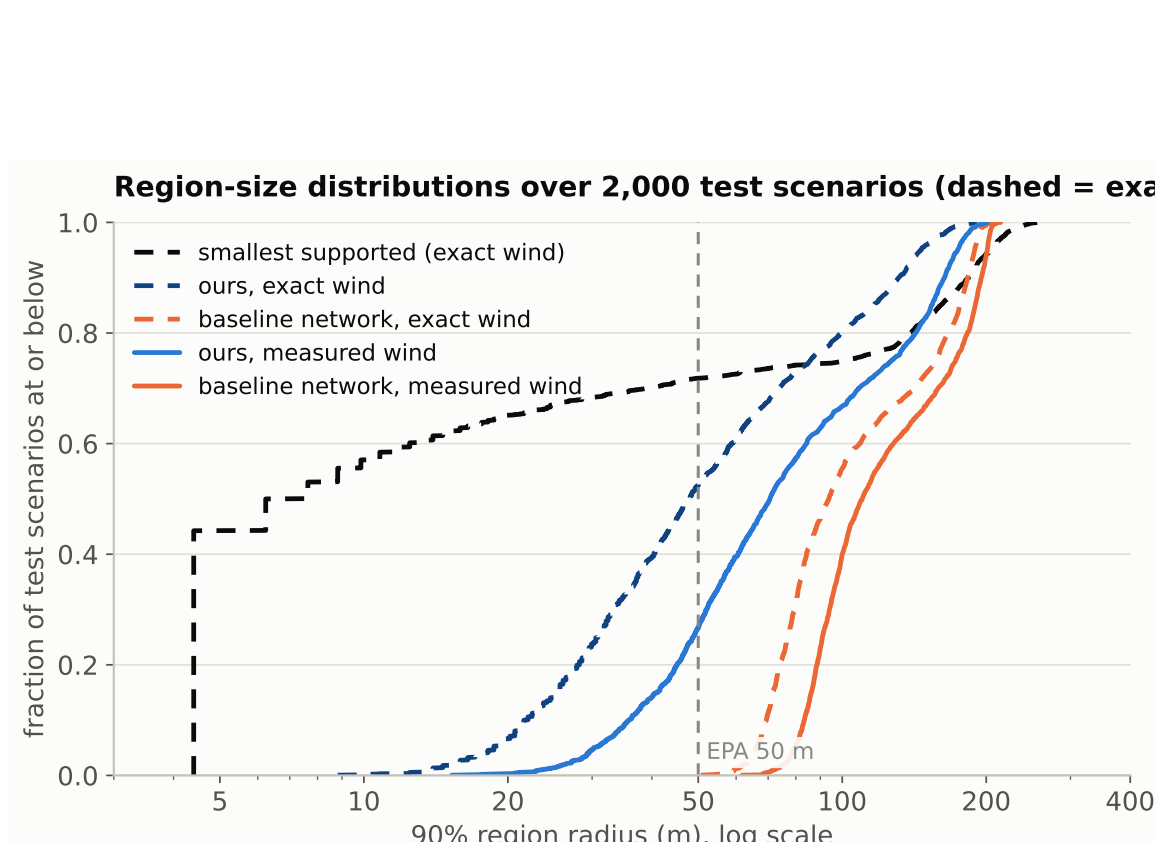
Median 90% search regions, drawn to scale on the 500 m site



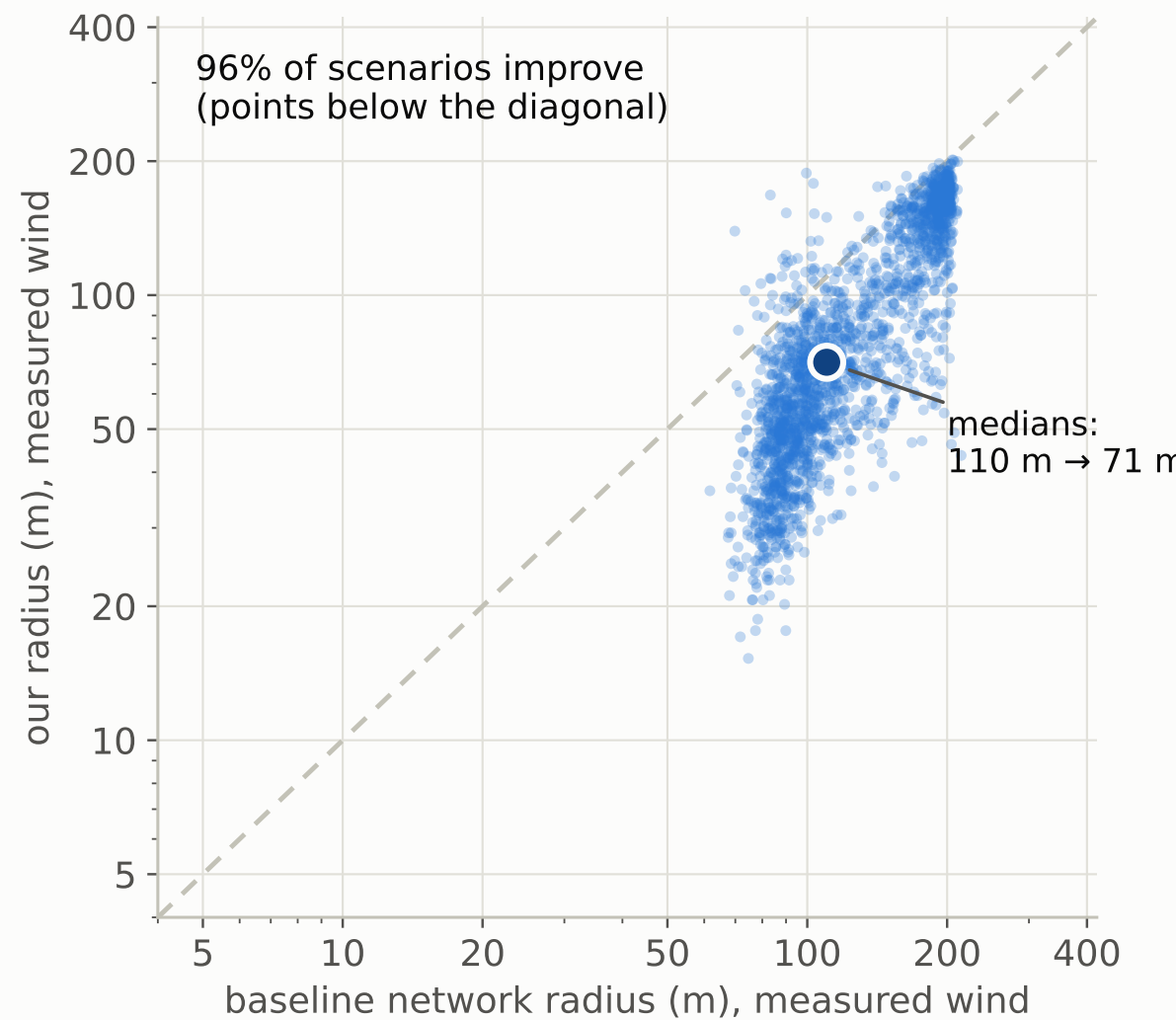
The physics images move every estimator into the same narrow band

Measured wind ($10^\circ/10\%$ error): each added physics image tightens the guaranteed region

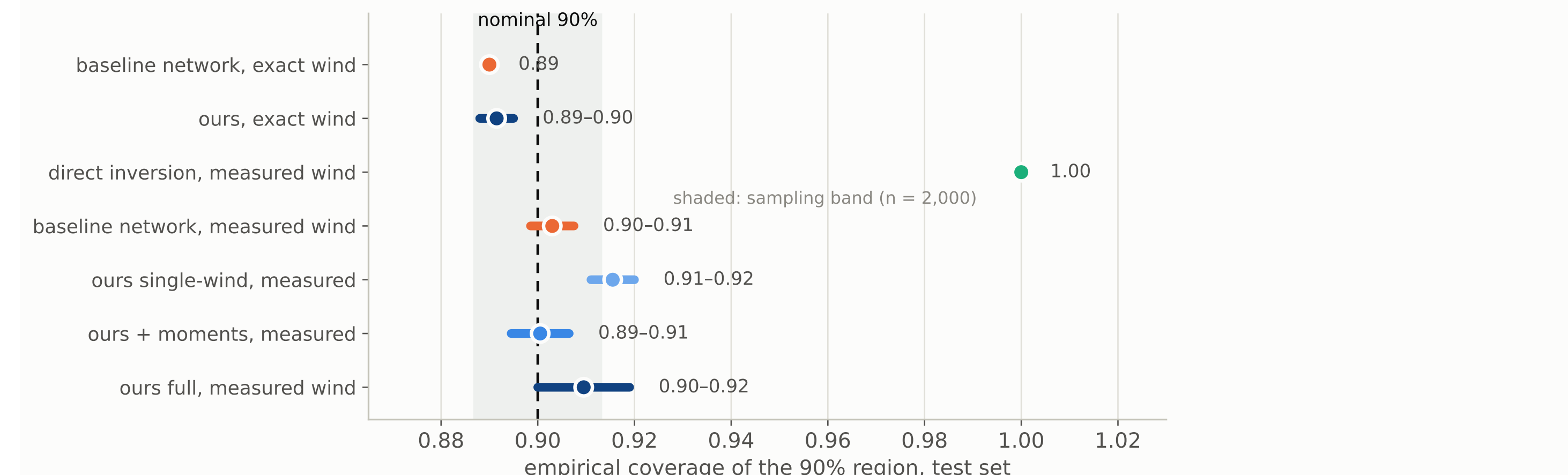
Region-size distributions over 2,000 test scenarios (dashed = exact wind, solid = measured wind)



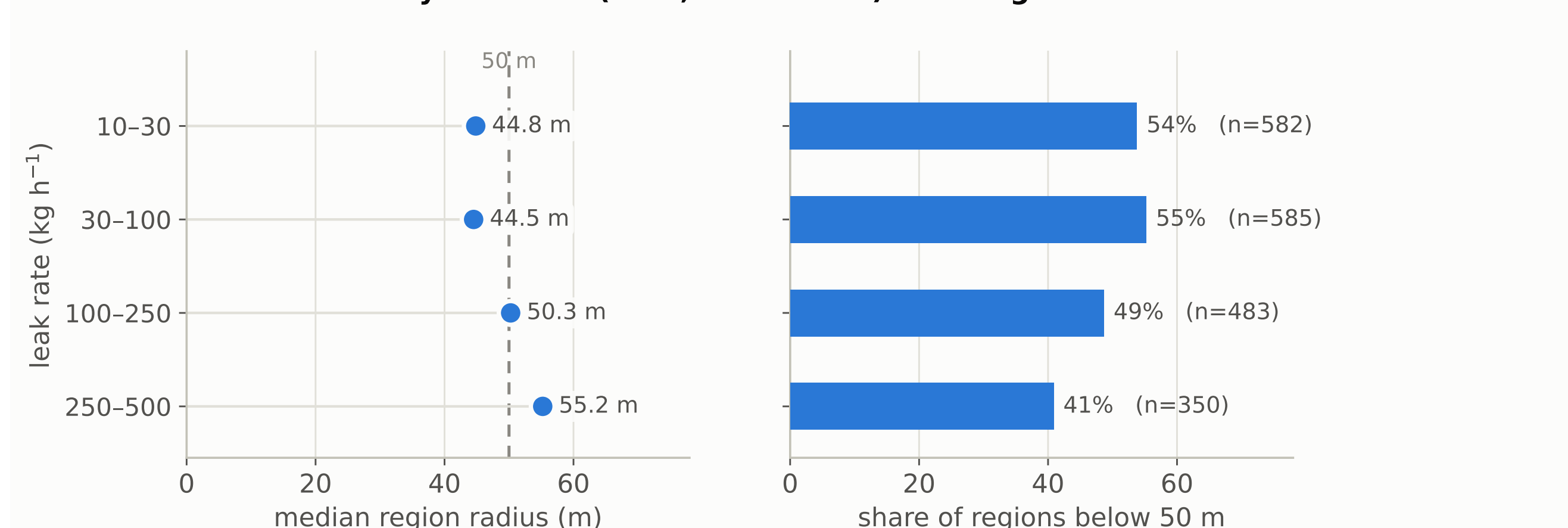
Scenario-by-scenario, same test set



The guarantee holds everywhere; direct inversion holds it only at coverage 1.0 (the full site)



Performance by leak rate (ours, exact wind): the largest emitters are the hardest cases



Discussion / Conclusions

Calibrated is not actionable. Both learned models pass every calibration check; only the exact-posterior audit shows one delivers regions $10\times$ larger than the data support.

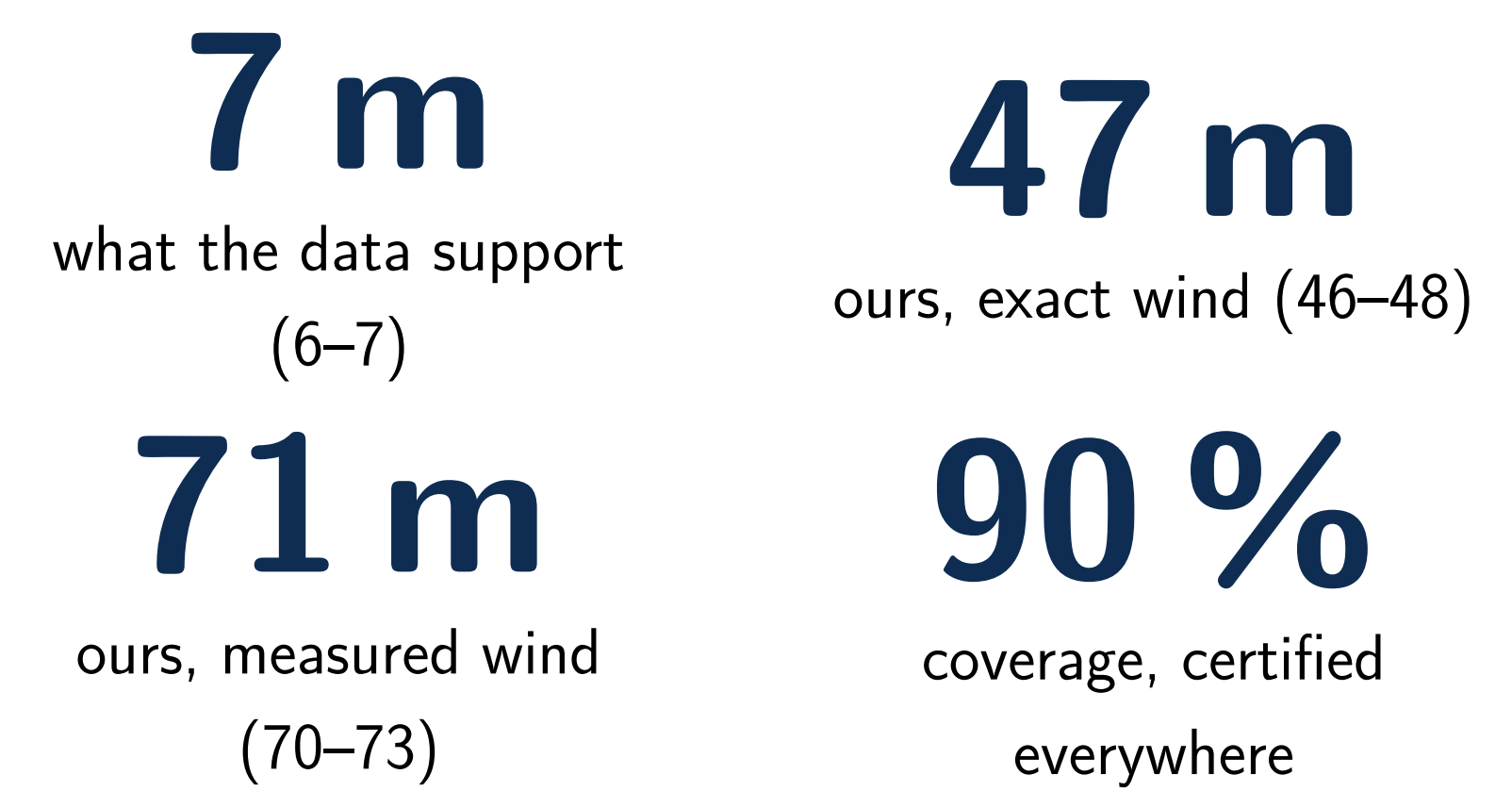
The fix belongs in the inputs. Supplying the classical evidence computation as images moves DeepSets, GNN, and Set Transformer from a 40 m spread into a **1.5 m band at 46–48 m**; the same images improve a flow-based estimator, so the recipe composes with any architecture.

Wind uncertainty, budgeted honestly. Under measured wind, direct inversion keeps its guarantee only by returning the full site. Each image of wind-error structure tightens the learned regions: $108 \rightarrow 82 \rightarrow 75 \rightarrow 71$ m, a quarter of cases below 50 m, at one forward pass.

The decomposition. $94 \rightarrow 47$ m: the missing computation, fixed by inputs. $47 \rightarrow 71$ m: the price of imperfect wind. $108 \rightarrow 71$ m: the value of physics when the wind is real.

Limitations & next steps. Simulation testbed (single steady source, flat terrain); error budget assumed known; largest emitters hardest. Next: misspecified budgets, controlled-release field validation (METEC-style).

At a Glance



References

- [1] U.S. EPA, NSPS 0000b/EG 0000c & Super-Emitter Program, 2024.
- [2] UNEP IMEO, *An Eye on Methane*, 2024.
- [3] Angelopoulos & Bates, *Conformal Prediction*, FNTML 2023.
- [4] Papamakarios & Murray, NeurIPS 2016.
- [5] Cannon et al., *Model Misspecification in SBI*, 2022.
- [6] Ilonze et al., *Elementa* 2025 (METEC).

Acknowledgements

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